Arm-level models for ORR and median PFS

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Data

Each row is one trial arm (or one non-overlapping stratum of an arm) from the `claude_collected` sheet of `data/raw/step3_stratum_arm_variables.xlsx`: 199 arms from 120 trials. Arms missing any covariate or the outcome are excluded (complete cases), so each model uses fewer arms (see the fit summaries).

Table 1: Covariates $\mathbf{x}_i$ (number of arms missing each)

Symbol	Covariate	Definition	Missing
age_i	Median age (years)	median age	21
F_i	Female proportion	female / n_i	18
T_i	Ever-smoker proportion	ever / (ever + never), known smoking status only	60
E_i	ECOG 0 proportion	ECOG 0 / n_i	43
S_i	Squamous proportion	squamous / n_i	35
G_i	Stage IV proportion	stage IV / stage I–IV, known stage only	55
L_i	First-line (0/1)	1 = first-line incl. maintenance; 0 = pretreated	0
$M_{i,k}$	MOA components (0/1 each)	chemo, ICI, other immune, anti-angio, targeted, other; all 0 = no active drug	0
B_i	Matched biomarker (0/1)	1 = selected on a validated predictive biomarker for a drug in the regimen	0

$\mathbf{x}_i = (\text{age}_i, F_i, T_i, E_i, S_i, G_i, L_i, M_{i,1}, \ldots, M_{i,6}, B_i)^\top$; the intercept is kept separate. n_i is the arm sample size.

Model 1: logistic models for ORR

Let $Y_i = \text{round}(\text{ORR}_i \cdot n_i)$ be the number of responders in arm i. All three versions share the mean model

$$\text{logit}(p_i) = \alpha_0 + \alpha^\top \mathbf{x}_i,$$

where p_i is the response probability of arm i and $\exp(\alpha_k)$ is the odds ratio of response per unit increase in covariate k. They differ only in the variance of Y_i.

Model 1.1, binomial. $Y_i \sim \text{Binomial}(n_i, p_i)$, with log-likelihood

$$\ell(\alpha_0, \alpha) = \sum_{i=1}^{N} \left[Y_i \lambda_i - n_i \log\left(1 + e^{\lambda_i}\right) \right] + \text{const}, \qquad \lambda_i = \alpha_0 + \alpha^\top \mathbf{x}_i.$$

Model 1.2, quasibinomial. Same mean, $\text{Var}(Y_i) = \phi\, n_i p_i (1 - p_i)$. The point estimates equal Model 1.1; every standard error is multiplied by $\sqrt{\hat{\phi}}$, where $\hat{\phi}$ is the Pearson χ^2 divided by the residual degrees of freedom.

Model 1.3, beta-binomial. $Y_i \mid q_i \sim \text{Binomial}(n_i, q_i)$ with $q_i \sim \text{Beta}(p_i s, (1 - p_i)s)$, so $\text{Var}(Y_i) = n_i p_i (1 - p_i)[1 + (n_i - 1)\rho]$ with $\rho = 1/(1 + s)$. Likelihood

$$L(\alpha_0, \alpha, s) = \prod_{i=1}^{N} \binom{n_i}{Y_i} \frac{B(Y_i + p_i s,\ n_i - Y_i + (1 - p_i)s)}{B(p_i s,\ (1 - p_i)s)},$$

maximized numerically; standard errors from the inverse Hessian.

Fit summary

Table 2: Model 1 fit summary

Model	Arms	Parameters	Dispersion	logLik	AIC
1.1 Binomial	84	15	Pearson $\chi^2/\text{df} = 8.54$	-477.1	984.3
1.2 Quasibinomial	84	16	$\hat{\phi} = 8.54$	–	–
1.3 Beta-binomial	84	16	$\hat{\rho} = 0.050$	-313.8	659.6

The binomial variance clearly fails: the Pearson ratio is far above 1, so Model 1.1 standard errors are too small by about $\sqrt{\hat{\phi}}$. The beta-binomial improves the log-likelihood by 163 units with one extra parameter.

Coefficients

Table 3: Model 1.1, binomial

Term	Estimate	SE	OR (95% CI)	p
Intercept	-6.136	0.649	0.00 (0.00, 0.01)	3.3e-21
Median age (years)	0.042	0.011	1.04 (1.02, 1.07)	1.2e-04
Female proportion	-0.213	0.293	0.81 (0.46, 1.43)	0.466
Ever-smoker proportion	-1.583	0.187	0.21 (0.14, 0.30)	2.7e-17
ECOG 0 proportion	-1.079	0.320	0.34 (0.18, 0.64)	7.6e-04
Squamous proportion	0.075	0.122	1.08 (0.85, 1.37)	0.540
Stage IV proportion	2.226	0.434	9.26 (3.96, 21.67)	2.9e-07
First-line	1.153	0.065	3.17 (2.79, 3.60)	1.3e-69
MOA: chemotherapy	1.009	0.094	2.74 (2.28, 3.30)	1.2e-26
MOA: immune checkpoint inhibitor	0.767	0.066	2.15 (1.89, 2.45)	5.6e-31
MOA: other immune	-0.105	0.112	0.90 (0.72, 1.12)	0.349
MOA: anti-angiogenic	0.561	0.062	1.75 (1.55, 1.98)	1.9e-19
MOA: targeted	0.927	0.093	2.53 (2.11, 3.03)	2.0e-23
MOA: other agent	0.334	0.097	1.40 (1.15, 1.69)	5.8e-04
Matched biomarker	0.437	0.114	1.55 (1.24, 1.93)	1.2e-04

Table 4: Model 1.2, quasibinomial

Term	Estimate	SE	OR (95% CI)	p
Intercept	-6.136	1.897	0.00 (0.00, 0.09)	0.001
Median age (years)	0.042	0.032	1.04 (0.98, 1.11)	0.189
Female proportion	-0.213	0.855	0.81 (0.15, 4.32)	0.803
Ever-smoker proportion	-1.583	0.547	0.21 (0.07, 0.60)	0.004
ECOG 0 proportion	-1.079	0.936	0.34 (0.05, 2.13)	0.249
Squamous proportion	0.075	0.357	1.08 (0.54, 2.17)	0.834
Stage IV proportion	2.226	1.267	9.26 (0.77, 111.00)	0.079
First-line	1.153	0.191	3.17 (2.18, 4.61)	1.6e-09
MOA: chemotherapy	1.009	0.276	2.74 (1.60, 4.71)	2.5e-04
MOA: immune checkpoint inhibitor	0.767	0.194	2.15 (1.47, 3.15)	7.5e-05
MOA: other immune	-0.105	0.327	0.90 (0.47, 1.71)	0.749
MOA: anti-angiogenic	0.561	0.182	1.75 (1.23, 2.50)	0.002
MOA: targeted	0.927	0.272	2.53 (1.48, 4.31)	6.4e-04
MOA: other agent	0.334	0.284	1.40 (0.80, 2.44)	0.239
Matched biomarker	0.437	0.332	1.55 (0.81, 2.97)	0.188

Table 5: Model 1.3, beta-binomial

Term	Estimate	SE	OR (95% CI)	p
Intercept	-7.032	1.655	0.00 (0.00, 0.02)	2.1e-05
Median age (years)	0.028	0.023	1.03 (0.98, 1.08)	0.223
Female proportion	-0.779	0.770	0.46 (0.10, 2.08)	0.312
Ever-smoker proportion	-1.368	0.461	0.25 (0.10, 0.63)	0.003
ECOG 0 proportion	0.438	0.674	1.55 (0.41, 5.80)	0.516
Squamous proportion	0.049	0.326	1.05 (0.55, 1.99)	0.881
Stage IV proportion	3.992	1.234	54.15 (4.82, 607.91)	0.001
First-line	0.949	0.173	2.58 (1.84, 3.63)	4.1e-08
MOA: chemotherapy	0.863	0.226	2.37 (1.52, 3.69)	1.3e-04
MOA: immune checkpoint inhibitor	0.562	0.193	1.75 (1.20, 2.56)	0.004
MOA: other immune	-0.343	0.301	0.71 (0.39, 1.28)	0.255
MOA: anti-angiogenic	0.614	0.163	1.85 (1.34, 2.54)	1.6e-04
MOA: targeted	0.683	0.242	1.98 (1.23, 3.18)	0.005
MOA: other agent	0.189	0.407	1.21 (0.54, 2.68)	0.643
Matched biomarker	0.926	0.281	2.52 (1.45, 4.38)	9.9e-04

Comparison

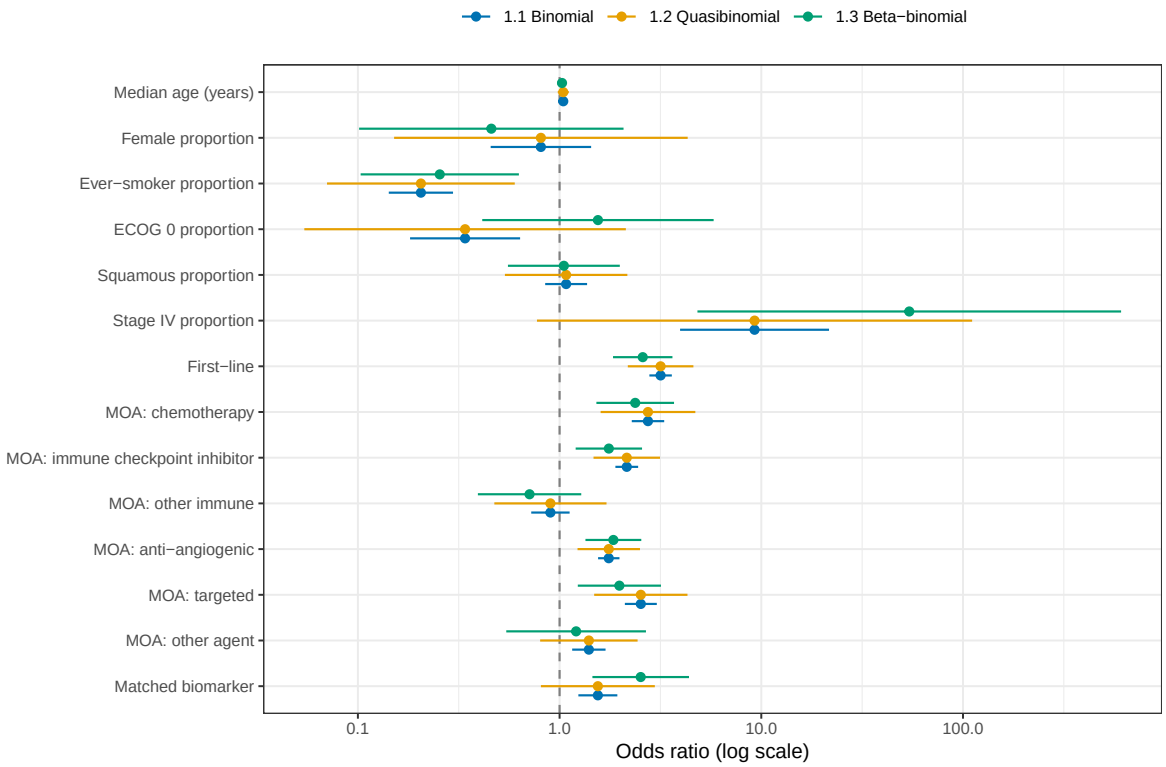


Figure 1: Odds ratios with 95% CIs by model (log scale), intercept omitted.

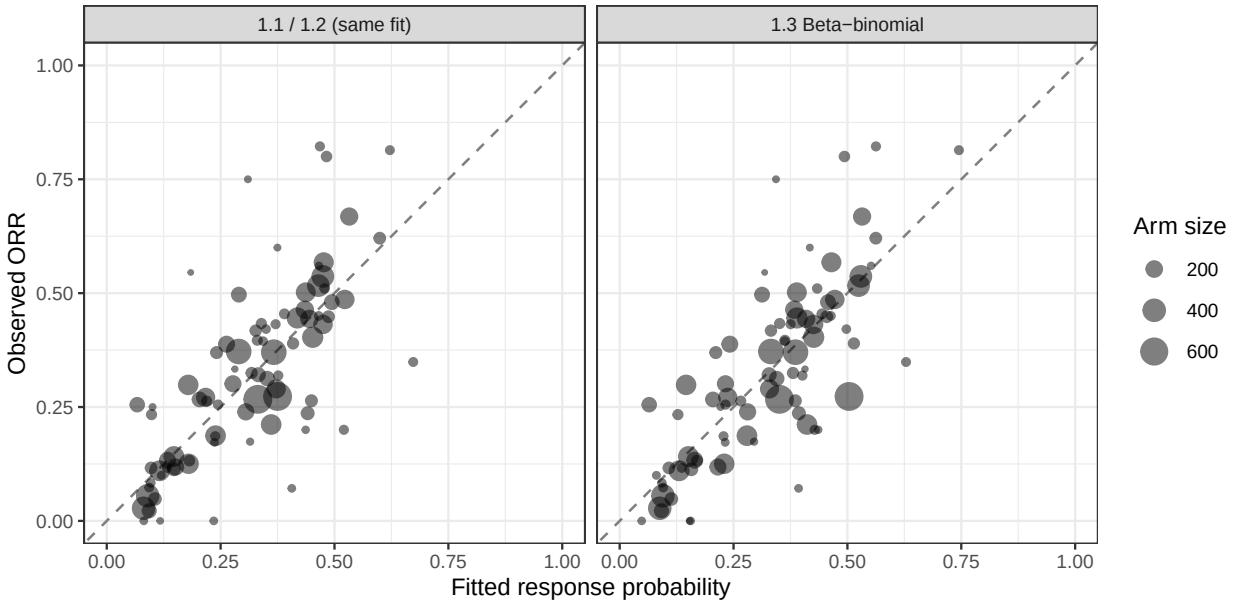


Figure 2: Observed ORR against fitted $\hat{p}_i$; point size is arm size. Points should scatter around the diagonal.

Model 2: log-linear model for median PFS

Let m_i be the reported median PFS (months) of arm i:

$$\log m_i = \theta_0 + \theta_p p_i + \theta^\top \mathbf{x}_i + \varepsilon_i, \qquad \varepsilon_i \sim N(0, \sigma^2/n_i),$$

fitted by weighted least squares with weights n_i. $\exp(\theta_k)$ is the ratio of median PFS per unit increase in covariate k. The response probability p_i is not observed and is either

- **observed:** the observed ORR_i (measurement error biases θ_p toward 0), or
- **estimated:** $\hat{p}_i$ from Model 1.1 (two-stage); standard errors and CIs from 1000 bootstrap resamples of arms.

Observed ORR

Table 6: Model 2 with observed ORR (80 arms, residual SE 2.07)

Term	Estimate	SE	Median ratio (95% CI)	p
Intercept	-0.788	0.597	0.45 (0.14, 1.50)	0.191
Response probability p_i	1.261	0.200	3.53 (2.37, 5.26)	2.9e-08
Median age (years)	0.031	0.010	1.03 (1.01, 1.05)	0.003
Female proportion	0.035	0.276	1.04 (0.60, 1.80)	0.899
Ever-smoker proportion	-0.169	0.191	0.84 (0.58, 1.24)	0.380
ECOG 0 proportion	0.150	0.297	1.16 (0.64, 2.10)	0.615
Squamous proportion	-0.150	0.122	0.86 (0.67, 1.10)	0.225
Stage IV proportion	-0.308	0.397	0.74 (0.33, 1.62)	0.441
First-line	0.175	0.075	1.19 (1.03, 1.38)	0.023
MOA: chemotherapy	0.419	0.091	1.52 (1.27, 1.82)	1.9e-05
MOA: immune checkpoint inhibitor	0.134	0.070	1.14 (0.99, 1.31)	0.060
MOA: other immune	-0.032	0.107	0.97 (0.78, 1.20)	0.766
MOA: anti-angiogenic	0.176	0.063	1.19 (1.05, 1.35)	0.007
MOA: targeted	0.087	0.092	1.09 (0.91, 1.31)	0.349
MOA: other agent	0.010	0.115	1.01 (0.80, 1.27)	0.928
Matched biomarker	0.423	0.105	1.53 (1.24, 1.88)	1.5e-04

Estimated $\hat{p}_i$ (two-stage)

Table 7: Model 2 with $\hat{p}_i$ from Model 1.1 (80 arms)

Term	Estimate	SE	Median ratio (bootstrap 95% CI)	p
Intercept	-1.781	1.898	0.17 (0.00, 5.97)	0.348
Response probability p_i	-0.472	2.030	0.62 (0.01, 38.49)	0.816
Median age (years)	0.046	0.028	1.05 (0.99, 1.11)	0.103
Female proportion	-0.160	0.744	0.85 (0.14, 2.57)	0.830
Ever-smoker proportion	-0.708	0.721	0.49 (0.11, 1.92)	0.326
ECOG 0 proportion	-0.158	0.645	0.85 (0.28, 3.53)	0.806
Squamous proportion	-0.145	0.250	0.86 (0.50, 1.43)	0.561
Stage IV proportion	0.225	1.237	1.25 (0.10, 15.20)	0.856
First-line	0.543	0.417	1.72 (0.65, 3.64)	0.193
MOA: chemotherapy	0.742	0.392	2.10 (1.00, 4.87)	0.058
MOA: immune checkpoint inhibitor	0.407	0.332	1.50 (0.84, 3.12)	0.221
MOA: other immune	-0.060	0.236	0.94 (0.48, 1.26)	0.798
MOA: anti-angiogenic	0.370	0.265	1.45 (0.94, 2.63)	0.162
MOA: targeted	0.454	0.444	1.57 (0.61, 3.72)	0.307
MOA: other agent	0.122	0.175	1.13 (0.83, 1.67)	0.484
Matched biomarker	0.540	0.472	1.72 (1.13, 6.26)	0.253

θ_p is weakly identified here: $\hat{p}_i = \text{logit}^{-1}(\hat{\alpha}_0 + \hat{\alpha}^\top \mathbf{x}_i)$ is a function of the same $\mathbf{x}_i$ that enters Model 2, so θ_p is separated from θ only by the curvature of the logit. The wide interval for θ_p reflects this, not a weak ORR–PFS

association. A joint model with latent p_i informed by the observed responders is the principled alternative.

Notes

- ORR is reported as a proportion; Y_i is reconstructed by rounding $\text{ORR}_i \cdot n_i$.
- Arms from the same trial are treated as independent. A trial-level random effect would account for their correlation.
- Overlapping or post-treatment subgroups (NCT01523587 mainland China subgroup, NCT03850444 KEYNOTE-042 China study, NCT01774721 skin-AE subgroups, NCT01038661 maintenance-phase arms) were moved to the `irrelevant_arms` sheet before fitting.